

TILLABERI, Niger

WMO: 610360

Lat: 14.200N

Lon: 1.450E

Elev: 210

StdP: 98.83

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	17.8	-8.1	1.9	29.5	-7.6	2.0	29.1	8.9	25.7	7.7	25.5	0.6	40	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.8	43.2	22.0	42.2	21.7	41.4	21.7	26.7	34.2	26.3	33.9	26.0	33.6	2.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.8	29.5	24.6	20.1	29.3	24.2	19.6	29.1	84.9	34.2	83.2	33.6	81.8	33.5	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.7	4.9	13.4	44.9	1.6	1.2	12.3	45.7	11.3	46.4	10.4	47.1	9.3	47.9		
			6.7	29.3	0.8	1.7	6.1	30.6	5.6	31.6	5.1	32.5	4.5	33.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	25.2	28.1	31.6	34.6	35.3	33.2	30.4	29.1	30.5	32.1	29.6	26.2	(6)
(7)		DBStd	3.61	2.18	2.75	2.28	1.83	1.99	2.32	2.22	1.80	1.96	1.45	1.69	1.98	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	7486	472	508	669	739	783	695	632	593	617	685	589	503	(10)
(11)		CDD18.3	4444	214	275	411	489	525	445	373	335	367	427	339	245	(11)
(12)		CDH23.3	63802	2211	3501	6257	8181	8771	7000	5123	4166	5008	6321	4587	2677	(12)
(13)		CDH26.7	39655	1042	2014	4093	5843	6309	4656	2838	1990	2767	3962	2779	1361	(13)
(14)	Wind	WSAvg	1.7	1.9	2.1	2.0	1.6	2.0	2.2	1.8	1.4	1.2	1.0	1.2	1.5	(14)
(15)	Precipitation	PrecAvg	399	0	0	3	2	19	53	104	149	78	10	0	0	(15)
(16)		PrecMax	698	0	9	36	22	86	127	230	359	368	131	10	4	(16)
(17)		PrecMin	212	0	0	0	0	0	7	26	0	0	0	0	0	(17)
(18)		PrecStd	105	0	1	7	5	20	27	47	71	67	21	2	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	40.8	42.3	44.5	44.4	42.3	39.4	36.8	39.4	40.7	39.5	37.3	(19)
(20)			MCWB	16.9	19.2	19.2	21.5	22.4	23.9	24.7	25.8	24.9	22.2	19.6	18.0	(20)
(21)		2%	DB	35.0	38.8	41.3	43.3	43.2	41.1	38.1	35.6	37.9	39.6	38.6	35.7	(21)
(22)			MCWB	16.2	18.4	19.0	21.3	22.9	24.3	24.9	25.5	25.2	22.5	19.4	17.2	(22)
(23)		5%	DB	33.5	37.2	40.3	42.5	42.3	39.9	36.9	34.6	36.6	38.8	37.8	34.5	(23)
(24)			MCWB	15.7	17.5	18.7	21.0	23.1	24.4	24.8	25.5	25.3	22.5	19.1	16.6	(24)
(25)		10%	DB	32.0	35.6	39.2	41.6	41.3	38.7	35.5	33.6	35.5	37.9	36.8	33.2	(25)
(26)			MCWB	15.1	16.7	18.3	20.8	23.3	24.4	24.8	25.3	25.3	22.6	18.6	16.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	21.8	24.1	25.7	26.6	26.6	26.7	27.1	27.3	26.9	23.3	19.2	(27)
(28)			MADB	34.2	34.2	32.0	38.7	38.6	35.7	33.5	33.5	34.2	35.4	34.0	34.5	(28)
(29)		2%	WB	17.2	20.0	22.0	24.7	25.9	26.0	26.1	26.5	26.7	26.1	21.8	18.3	(29)
(30)			MADB	32.7	36.3	34.2	37.4	37.4	35.8	33.3	32.8	33.7	34.9	33.6	33.5	(30)
(31)		5%	WB	16.4	18.6	20.4	24.0	25.5	25.6	25.7	26.1	26.2	25.5	20.7	17.5	(31)
(32)			MADB	31.7	34.6	37.4	36.9	36.9	35.4	32.9	32.2	33.3	34.4	33.4	32.4	(32)
(33)		10%	WB	15.6	17.6	19.3	23.4	25.0	25.2	25.5	25.8	25.9	25.1	19.7	16.7	(33)
(34)			MADB	30.6	33.7	36.9	36.4	36.5	35.0	32.5	31.6	32.9	33.8	33.3	31.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.7	14.3	14.7	14.3	11.8	10.4	9.0	8.3	9.5	12.3	15.0	14.1	(35)
(36)			MADB	15.8	16.4	16.2	15.4	13.2	11.5	10.5	9.6	11.0	13.5	15.7	15.7	(36)
(37)			MCWBR	5.6	5.4	5.0	4.7	3.5	2.5	2.3	2.5	2.6	3.6	4.5	5.2	(37)
(38)		5% WB	MADB	14.9	15.4	14.3	12.4	11.2	10.5	9.3	8.8	9.7	11.1	14.7	15.7	(38)
(39)			MCWBR	5.9	5.5	5.3	4.3	3.3	2.8	2.8	3.1	3.0	3.2	4.9	5.5	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub		0.583	0.622	0.712	0.737	0.743	0.724	0.660	0.613	0.596	0.635	0.559	0.542	(41)
(42)		taud		1.717	1.595	1.442	1.484	1.483	1.535	1.698	1.784	1.805	1.682	1.735	1.774	(42)
(43)		Ebn at Noon		717	712	663	648	633	637	682	723	737	693	731	737	(43)
		Edh at Noon		229	269	321	307	302	283	242	224	220	243	223	211	(43)
(44)	All-Sky Solar Radiation	RadAvg		5.54	6.07	6.42	6.57	6.48	6.10	5.96	5.81	6.19	6.04	5.70	5.37	(44)
(45)		RadStd		0.22	0.27	0.32	0.30	0.17	0.26	0.13	0.22	0.17	0.18	0.13	0.17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	+0.37	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page